Action-Conditioned Predictive Consistency for Robust Visual World Models

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Abstract

Latent predictive world models such as JEPAs predict future representations rather than pixels, but this alone does not define visual robustness for control. We argue that robustness should be measured as *action-conditioned predictive consistency*: unperturbed and corrupted views of the same state may encode differently, but under the same history and action sequence they should induce consistent task-relevant future predictions, while genuinely action-relevant state differences remain discriminable. We study this requirement on LeWorldModel (LeWM) across PushT, TwoRoom, Reacher, and Cube using 36 checkpoints and a unified 3-seed $\times$ 100-trajectory protocol. Under primary observation-only Gaussian noise with an unperturbed goal, a no-noise LeWM checkpoint drops from 86.33% to 4.33% on PushT and from 58.67% to 18.33% on Reacher. Input-side noise augmentation recovers much of this performance, but behaves as a coarse global pressure with broad, task-dependent plateaus. A paired Gaussian-noise ACPC-basin diagnostic shows that robust checkpoints produce much tighter same-action prediction basins (PushT R_F: 1.543 $\rightarrow$ 0.088), while cross-checkpoint analysis shows that pointwise fragility does not isolate the corruption gap after conditioning on train-time noise. A natural target-view ablation is negative: perturbed-history $\rightarrow$ original-future prediction reaches only 8.33% on PushT pixels 0.08 versus 89.00% for full-sequence perturbed targets. PLDM replication preserves the task-level signature, and heteroscedastic reweighting shows why "hard" transitions cannot be treated as nuisance. The paper contributes a reframing and diagnostics, not a new training algorithm.

Keywords: visual world models; JEPA; action-conditioned predictive consistency; visual robustness; predictive-dynamics diagnostics.

1 Introduction

1.1 Latent prediction is not a robustness definition

JEPAs predict future representations instead of reconstructing pixels, which can reduce pressure to encode high-magnitude irrelevant visual detail [2, 4, 35]. For control, however, "irrelevant" is action- and transition-dependent. A representation may ignore a background texture but must preserve a contact pose or doorway state if that changes the next useful action. Thus encoder-level invariance is the wrong object: planning uses the model's *predictions*, so robustness must be stated after action-conditioned prediction.

Our controlled probe makes this failure concrete. A no-noise LeWM [23] checkpoint scores 86.33% on unperturbed PushT but only 4.33% under observation-only Gaussian noise at std 0.08 with the goal unperturbed; Reacher drops from 58.67% to 18.33%, and TwoRoom from 94.00% to 65.67%. Pixels+goal corruption is reported as a harder auxiliary stress condition. Visual-corruption fragility is not new [39, 38, 15, 24]; our focus is what it reveals about JEPA latent world-model control.

1.2 From visual invariance to action-conditioned predictive consistency

We reframe visual robustness for world-model control at the level of *action-conditioned predictive dynamics*. Let $z = E(h)$ and $\tilde{z} = E(\tilde{h})$ be the latents an encoder E produces from an unperturbed history h and a corrupted history $\tilde{h}$ of the same underlying state, and let $F^k(\cdot, \mathbf{a}_{0:k-1})$ be the action-conditioned latent predictor rolled out for k steps under an action sequence $\mathbf{a}$. We *allow* $z \neq \tilde{z}$. The robustness target is placed *after* the predictor: in task-relevant coordinates,

$$F^k(z, \mathbf{a}_{0:k-1}) \;\approx\; F^k(\tilde{z}, \mathbf{a}_{0:k-1}), \qquad k = 1, \dots, H, \tag{1}$$

so that the two branches induce the same next-state and rollout predictions under the same action intervention. Crucially this is *selective*: it should hold only for nuisance perturbations of the same underlying state. State differences that change action-conditioned transitions, costs, or the optimal action must remain *discriminable*, or the planner loses the distinctions it needs. We make both halves precise in Section 3.

This reframing changes what counts as a robustness measurement. Encoder-level latent closeness, $\|z - \tilde{z}\| \to 0$, is neither necessary (a large but task-irrelevant encoder shift can wash out after the predictor) nor sufficient (a small encoder shift can still flip the planned action). Planning, cost, and action metrics remain useful, but as *downstream readouts* of predictive consistency rather than as the central definition; the object we ultimately want to regularise is the action-conditioned predictive dynamics.

1.3 Existing corruption results as a controlled probe

A natural remedy is input-side noise augmentation. We sweep `std_max` $\in \{0.01, \dots, 0.08\}$ across four tasks and find recovery, but not a single optimal setting: TwoRoom benefits from heavy noise, PushT's unperturbed and noisy-eval optima dissociate, Reacher has a broad plateau, and Cube responds weakly. This is the signature of a coarse global pressure, not a selective objective that knows which visual variation is nuisance and which controls future dynamics. We then run the natural target-view ablation, perturbing the input/history while keeping the predictive target at the original future view. It does not reproduce the robust recovery of full-sequence augmentation, so the method implication is narrower: target-view replacement alone is insufficient; paired predictive-dynamics consistency is the next training object. We do not present a new training algorithm.

1.4 Contributions

C1 — Problem reframing. We formulate visual robustness for world-model control as action-conditioned predictive consistency plus an action-relevant discriminability countercondition, rather than encoder-level invariance.

C2 — Diagnostic evidence. Across 36 LeWM checkpoints and a PLDM replication, we show severe corruption fragility, task-dependent recovery under input-side noise augmentation, and the limits of pointwise diagnostics: after conditioning on `std_max`, the strongest single-step fragility ratio does not explain the PushT corruption gap (partial Spearman $\rho = +0.19$, CI including 0).

C3 — Paired consistency probes. We report a dense Gaussian-noise ACPC basin diagnostic on all 36 LeWM checkpoints and keep the broader Phase-0 ACPC/PCC/CRA/MAF/ADM/SPRR sweep in Appendix H as exploratory evidence, not as a universal robustness predictor.

C4 — Method-design implication. We report the target-view ablation that the ACPC framing makes natural: full-sequence perturbed targets versus perturbed-history → original-future prediction. The latter is a negative ablation, sharply underperforming the full-sequence branch on robust evaluation. This rules out a naive target-view-only implementation and motivates paired predictive-dynamics consistency gated by action/transition sensitivity rather than prediction error.

2 Related Work

2.1 JEPA and latent world models

JEPA [21] predicts in latent space rather than reconstructing pixels; I-JEPA [2] and V-JEPA [4, 3] extend this idea to images and video. LeWM [23], our baseline, stabilises an end-to-end JEPA world model with SIGReg [7]. Its Violation-of-Expectation result probes surprise under physical and colour changes, whereas we measure control success under pixel corruptions. PLDM [29, 30], via `stable-worldmodel` [22], supplies the second method family in Appendix F; the main diagnostic analysis remains LeWM-centred.

2.2 Joint-embedding, reconstruction, and augmentation alignment

Augmentation and joint-embedding losses impose invariances only when their training signal identifies the right nuisance factors [37, 32, 41]. Van Assel et al. [35] show that joint-embedding prediction can reduce pressure to encode high-magnitude irrelevant features, but still needs augmentation or bias to decide *which* variations are irrelevant. In control, that decision is action- and transition-conditioned. Seq-JEPA [11] addresses a related invariance–equivariance tension architecturally; we instead state the requirement on predictions.

2.3 LeJEPA identifiability and latent world-model theory

Klindt et al. [18] analyse when LeJEPA recovers latent world variables and supports latent planning. Our question is complementary: given a learned visual world model, when do unperturbed and corrupted views of the same state induce consistent action-conditioned predictions? We do not claim an identifiability result.

2.4 Bisimulation and control-relevant representations

Bisimulation merges states with equivalent reward and transition behaviour. DeepMDP [10], DBC [40], bisimulation-regularized MPC [28], and Bisim-JEPA [34] use this principle to learn control-relevant representations. Our framing differs in that paired unperturbed/corrupted views of the *same* state need not share an encoder latent; we ask for agreement after the action-conditioned predictor while explicitly preserving action-distinct transitions.

2.5 Visual robustness in model-based and pixel-based RL

DrQ/DrQ-v2 [39, 38] and SODA/DMC-GB [15] establish augmentation-based visual robustness for pixel RL. DreamerV3 [13], TD-MPC2 [14], and `stable-worldmodel` [22] study visual world-model control, but do not settle the JEPA predictive-consistency question. ViGMO [24] is closest: it studies unseen visual distractions and adds latent consistency. Our boundary is that ViGMO regularises generic encoder consistency, whereas we place consistency on action-conditioned rollouts and pair it with a discriminability guard. JEPA robustness work such as N-JEPA [25], VJEPA [16], and US-JEPA [26] is compatible with our result because signal recovery and closed-loop action optimisation test different properties.

2.6 Value-aware and value-equivalent model learning

Value-equivalent and value-aware models [12, 36] argue that a model should serve downstream control rather than fit all dynamics equally. VIBR [6] similarly favours view-invariant value functions over invariant representations. We adopt the downstream-consequence principle, but apply it to predictive dynamics: same-state views should agree after action-conditioned prediction, while different action-relevant transitions stay distinct.

2.7 Representation diagnostics and collapse analysis

We use standard representation diagnostics: effective rank [27, 17, 33], CKA [19], linear probes [1], and anti-collapse context from VICReg [5]. RankMe [9] motivates our cross-checkpoint question: after removing the sweep-level training-noise trend, does a label-free predictor-sensitivity diagnostic retain downstream control signal? The individual metrics are standard; our contribution is the control-facing protocol and its scope boundaries.

3 Action-Conditioned Predictive Consistency

This section defines action-conditioned predictive consistency and its discriminability countercondition. Section 3.2 then separates the main descriptive diagnostics from paired ACPC probes.

3.1 Setup and definitions

Let o_t be an unperturbed observation and $\tilde{o}_t = \tau(o_t)$ a visually corrupted view of the *same* underlying state (τ is additive Gaussian pixel noise here, but the definitions are corruption-agnostic). Let h_t and $\tilde{h}_t$ be the corresponding observation–action histories, E_θ the encoder, and F_θ the action-conditioned latent predictor. Fix an action sequence $\mathbf{a} = a_{t:t+H-1}$ that is *identical* on both branches, and write

$$z_t = E_\theta(h_t), \quad \tilde{z}_t = E_\theta(\tilde{h}_t), \qquad \hat{z}_{t+k} = F_\theta^k(z_t, \mathbf{a}_{0:k-1}), \quad \hat{\tilde{z}}_{t+k} = F_\theta^k(\tilde{z}_t, \mathbf{a}_{0:k-1}). \tag{2}$$

Let Π denote a *task-relevant predictive readout* — the full latent, a transition delta, goal-distance or cost features, or a learned projection — and let d be a distance on its range.

Action-conditioned predictive consistency. For a same-state perturbation pair under the shared action sequence,

$$\mathrm{ACPC}_H(o_t, \tilde{o}_t, \mathbf{a}) = \sum_{k=1}^{H} \alpha_k \, d\big(\Pi(\hat{z}_{t+k}), \, \Pi(\hat{\tilde{z}}_{t+k})\big), \qquad \alpha_k \geq 0. \tag{3}$$

Robustness is the requirement that ACPC_H be *small* — crucially *not* that the encoder distance $d(z_t, \tilde{z}_t)$ be small. Encoder closeness is neither necessary (a large but task-irrelevant encoder shift can wash out through F_θ and Π) nor sufficient (a small encoder shift can still change the predicted future enough to flip the planned action).

Action-relevant discriminability (countercondition). Consistency must be *selective*. Let $Y^H(s, \mathbf{a})$ denote a task-relevant future readout that is external to the learned predictor — simulator state, cost-to-go, inverse-dynamics label, contact/keyframe tag, or another diagnostic proxy available in the dataset. For two genuinely different states s_i, s_j (with latents z_i, z_j) that admit an action sequence under which this external future readout diverges,

$$\Delta^H_{\text{dyn}}(s_i, s_j, \mathbf{a}) = d_Y\big(Y^H(s_i, \mathbf{a}),\, Y^H(s_j, \mathbf{a})\big) > m, \tag{4}$$

the representation and predictions must keep them separable,

$$d\big(\Psi(z_i),\, \Psi(z_j)\big) > m', \tag{5}$$

where Ψ is a discriminability readout (latent, predicted delta, inverse-dynamics feature, cost feature, or rollout embedding) and $m, m' > 0$ are margins. Equations (3)–(5) make the central tension precise: it is not invariance versus resolution in the abstract, but

predictive consistency high for same-state visual-perturbation pairs; predictive discriminability high for state/action/transition-distinct pairs.

A method that drove $\text{ACPC}_H \to 0$ by collapsing the latent would violate (5), which is why the two conditions are stated together.

3.2 Operational consistency diagnostics

Status. Descriptive quantities support the main cross-checkpoint analysis; paired ACPC quantities define method-facing probes. The paper-facing paired result is the Gaussian-noise ACPC basin diagnostic on all 36 LeWM checkpoints (Section 5.5). Appendix H reports broader ACPC/PC-C/CRA/MAF/ADM/SPRR probes under the auxiliary pixels+goal stress condition as exploratory diagnostics. ϵ below is the constant from Equation (7).

Encoder perturbation difference (EPD), descriptive.
EPD $= d(E(o), E(\tilde{o}))$. This is the Layer-1 encoder shift of Section 4.3, retained only as a *descriptive* quantity: a large EPD is not in itself bad and a small EPD is not in itself good. It reports whether a perturbation enters the latent, not whether it matters for control.

One-step consistency (ACPC-1), paired probe.
$\text{ACPC}_1 = d\big(F(E(o), a), F(E(\tilde{o}), a)\big)$, optionally normalised by a natural transition magnitude, $\text{ACPC}_1/(d(z_t, z_{t+1}) + \epsilon)$, so that it is comparable across checkpoints.

Multi-step consistency (ACPC-H), paired probe.
Equation (3) with $H \in \{4, 8\}$. This is the principled, *paired* version of the existing multi-step rollout drift (`predictor_rollout_T8_l2`): it measures disagreement between unperturbed and corrupted rollouts under the *same* action sequence, not generic rollout drift.

ACPC basin radii, reported diagnostic.
For each state we evaluate the original view plus Gaussian-noised views at std $\in \{0.01, \dots, 0.08\}$. We report an encoder radius R_E (median pairwise encoder-view L2 spread, normalised by the original-view NN L2 scale), a prediction radius R_F (median pairwise rollout-view L2 spread, normalised by the original transition L2 scale), and a contraction ratio R_F/R_E. This is the dense paired diagnostic in `assets/paper1_data/acpc_basin_diagnostics.json`.

Predictive cost consistency (PCC), downstream readout.
PCC $= \big|J(F^H(E(o), \mathbf{a}), g) - J(F^H(E(\tilde{o}), \mathbf{a}), g)\big|$ for a cost/goal readout J and goal g. PCC tests whether predictive mismatch changes the planning cost; it is a downstream readout, not the central definition.

Candidate-ranking agreement (CRA), downstream readout.
For a candidate set $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\}$ held identical across branches, the Spearman/Kendall agreement of the unperturbed and corrupted cost vectors. We treat CRA as a downstream readout, require the candidate set to be shared, and do not call it planning equivalence.

Margin-conditioned action flip (MAF), downstream readout.
$\mathrm{MAF} = \mathbf{1}[u^*_{\mathrm{orig}} \neq u^*_{\mathrm{noise}},\ C_{(2)} - C_{(1)} > \delta]$: a flip of the selected action counts only when the unperturbed top-1/top-2 cost margin exceeds δ, since a flip inside the noise margin need not be a failure.

Action-relevant discriminability margin (ADM), paired probe.
Over action/transition-distinct pairs P_{disc} (differing inverse-dynamics labels, large original-view transition magnitude, contact/keyframe transitions, or large cost-to-go change), $\mathrm{ADM} = \mathrm{median}_{(i,j)\in P_{\mathrm{disc}}} d(\Psi_i, \Psi_j)$. A consistency method should reduce ACPC_H without materially reducing ADM.

Selective predictive robustness ratio (SPRR), summary probe.
A single summary of action-distinct discriminability relative to original/corrupted predictive inconsistency,

$$\mathrm{SPRR} = \frac{\mathrm{median}_{(i,j)\in P_{\mathrm{disc}}} d(\Psi_i, \Psi_j)}{\mathrm{median}_{P_{\mathrm{pert}}} \mathrm{ACPC}_H + \epsilon}.$$

We use it descriptively and do not claim it predicts success.

These diagnostics localise mechanisms and define method targets; consistent with the negative results in Section 5.7, we do not claim that any of them *predict* robustness.

4 Background and Diagnostic Framework

4.1 LeWorldModel baseline

LeWM [23] is an end-to-end JEPA world model trained with two terms only:

$$\mathcal{L}_{\mathrm{LeWM}} \;=\; \mathcal{L}_{\mathrm{pred}} \;+\; \lambda \cdot \mathcal{L}_{\mathrm{SIGReg}}\,, \tag{6}$$

where $\mathcal{L}_{\mathrm{pred}}$ is the latent-space mean-squared error (MSE) between predicted and target representations. SIGReg projects each latent onto M unit-norm random directions, computes the Epps–Pulley empirical-characteristic-function distance [7] between each projection and $\mathcal{N}(0,1)$, and aggregates with Gauss-window weights, preventing collapse without explicit BatchNorm. (The Cramér–Wold theorem motivates the construction: equality of high-dimensional distributions reduces to equality of all one-dimensional projections of their characteristic functions.) Inference uses the Cross-Entropy Method (CEM) [20] for latent-space model predictive control (MPC) [8].

4.2 Input-side noise augmentation

We add per-frame Gaussian noise to the LeWM input pipeline via `utils.py::AddNormalizedGaussianNoise` (Appendix A). Each frame is independently noised: Bernoulli(p) decides whether the frame is corrupted, and if so the standard deviation is drawn from $\mathrm{Uniform}(0, \mathrm{std\_max})$. The main sweep applies the training transform to the whole `pixels` sequence, so the predictive target latent is also encoded from a perturbed future frame; this is a full-sequence augmentation pressure, *not* an original-target objective. We fix $p = 1.0$ and sweep `std_max` $\in \{0.01, \ldots, 0.08\}$ (eight levels). We also train a target-view ablation with the same perturbed input/history but an original, unperturbed future target. Our primary noised evaluation applies Gaussian noise only to the observation pixels while leaving the goal image unperturbed; pixels+goal evaluation is retained as a stronger full-visual-stream stress test.

4.3 Five-layer diagnostic framework

The framework decomposes the JEPA world-model control forward pass — pixels $\rightarrow$ encoder f $\rightarrow$ latent z $\rightarrow$ predictor g $\rightarrow$ cost surface $\rightarrow$ planned actions (optimised with CEM) — into five sequential stages and instruments each transition with one or more metrics. Layers 1–2 characterise the encoder output; Layer 3 measures how the predictor amplifies an encoder shift; Layer 4 injects noise directly into the latent to isolate predictor and cost contributions; Layer 5 quantifies planning-relevant information retained in the latent. Most individual metrics already have a literature home (cited inline below). Our additions are scale-normalised ratios that compare perturbation effects with the unperturbed local nearest-neighbour scale. We use nearest-neighbour distance as a local latent scale primitive in the spirit of k-nearest-neighbour (kNN) out-of-distribution detection [31], but the ratios themselves are introduced here.

Relation to action-conditioned predictive consistency. The five layers decompose input-side noise augmentation into encoder shift, encoder geometry, predictor sensitivity, latent-noise response, and task resolution. Layers 1–2 describe whether a perturbation enters the latent; Layer 3 is closest to predictive consistency through fragility ratio and rollout drift; Layer 5 measures the discriminability side through transition resolution and inverse-dynamics probes. We treat encoder closeness as descriptive only, because robustness is tested after the action-conditioned predictor. Figure 1 illustrates the boundary, and Table 11 summarises the task-level reading.

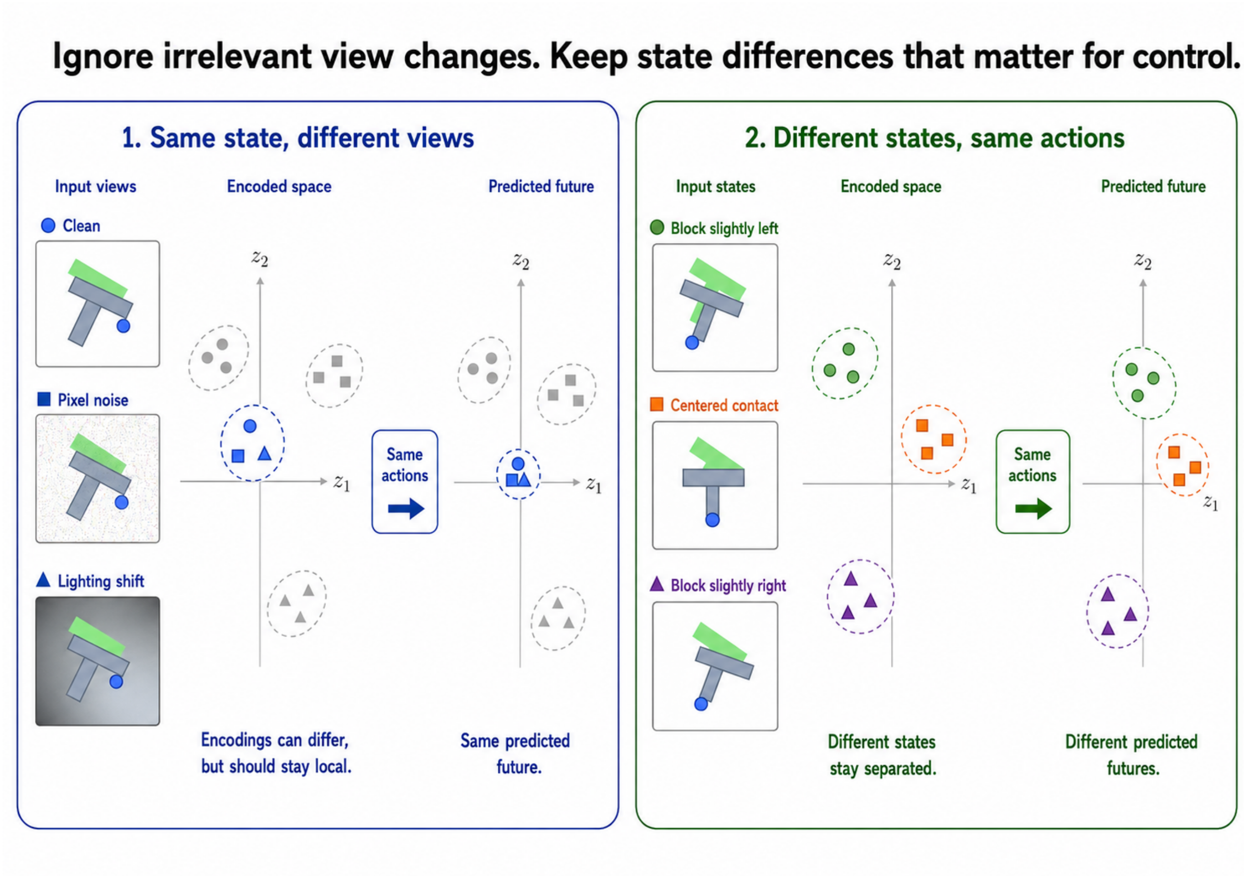

Figure 1: Conceptual reading of the selective-consistency tension. Same-state nuisance variants should induce consistent action-conditioned predictions; states that differ in transition, cost, or optimal action must remain resolvable. The bottom axis is a qualitative task read backed by the sweep and diagnostic evidence; it should not be read as "TwoRoom needs no resolution" or "PushT should preserve all pixels". PLDM is used as a second-family replication of the task-level signature, while the exact diagnostic mechanism remains method-specific. Section 3 formalises the boundary as action-conditioned predictive consistency: contract nuisance perturbations only after the action-conditioned predictor, while keeping action-distinct transitions resolvable.

Minimal notation. Let $z_i = f(x_i)$ and $\tilde{z}_i = f(\tilde{x}_i)$ be unperturbed and noised latents for the same token. The distance, local nearest-neighbour (NN) scale, and three introduced ratios are defined together as

$$\begin{aligned} d_{\cos}(a,b) &= 1 - \frac{a^\top b}{\|a\|\,\|b\|}, \\ d_i^{\mathrm{NN}} &= \min_{j\neq i} d_{\cos}(z_i, z_j), \\ r_i^{\mathrm{enc}} &= \frac{d_{\cos}(z_i,\tilde{z}_i)}{d_i^{\mathrm{NN}}+\epsilon}, \\ r_i^{\mathrm{pred}} &= \frac{d_{\cos}\big(g(z)_i, g(\tilde{z})_i\big)}{d_i^{\mathrm{NN}}+\epsilon}, \\ R_d &= \frac{\mathrm{median}_t\ d(z_t,z_{t+1})}{\mathrm{median}_{(t,t')\in\mathcal{F}}\ d(z_t,z_{t'})+\epsilon}. \end{aligned} \tag{7}$$

Here r_i^{enc} is the encoder-shift ratio, r_i^{pred} is the fragility ratio, and R_d is the transition-resolution ratio for either cosine or L2 distance d. $\mathcal{F}$ denotes random far pairs from different temporal locations, and $\epsilon = 10^{-8}$ is a numerical-stability constant. Released JSON artifacts keep the implementation keys (`noise_to_nn_cos_ratio`, `predictor_target_to_nn_cos_ratio_at_max_std`, and `transition_resolution_ratio_{cos,l2}`), but the main text uses the shorter prose names above.

Layer 1 — Encoder shift. The magnitude and direction of latent displacement under input noise. Metrics: raw angular/L2 shift, angular gain per unit noise, and the encoder-shift ratio in Equation (7).

Layer 2 — Encoder geometry. The global structure of the encoded latent space. Metrics: local neighbourhood scale, effective rank [27], and Centered Kernel Alignment (CKA) [19] between unperturbed-input and noisy-input latents at the diagnostic's maximum injection std.

Layer 3 — Predictor sensitivity. How the predictor amplifies the encoder shift. Metrics: the *fragility ratio* in Equation (7) and multi-step rollout L2 drift at horizon T.

Layer 4 — Latent-noise response. Noise injected directly into the latent z (bypassing the encoder) isolates predictor and cost contributions. Metrics: the slope of the planner's cost surface under latent perturbations and the latent robust radius, defined as the largest perturbation that keeps the cost ranking stable.

Layer 5 — Task resolution. The control-relevant information retained in the latent. Metrics: the L2/cosine transition-resolution ratio R_d in Equation (7) and the inverse-dynamics linear-probe R^2, a controllability proxy in the spirit of linear-probe analysis [1].

Reading guidance. Layers 1, 2, 4, and 5 are descriptive. Layer 3 supplies the two full-coverage cross-checkpoint metrics used in Section 5.7: fragility ratio and multi-step rollout drift.

4.4 Cross-checkpoint validation protocol

For each task we compute Spearman ρ across the 9 LeWM checkpoints (base + `std_max` $\in \{0.01, \ldots, 0.08\}$) and partial Spearman conditioned on `std_max`. The partial step removes the sweep-level training-noise trend, asking whether a diagnostic retains residual checkpoint-quality signal. We report 95% percentile bootstrap intervals ($B = 1000$, seed $= 42$); the bootstrap artifact is `assets/paper1_data/partial_corr_bootstrap_20260523.json`. PLDM replication is reported in Appendix F.

5 Experiments

5.1 Setup

Tasks. PushT (2D pushing), TwoRoom (2D navigation), Reacher (2D arm), and Cube (3D manipulation) form a heterogeneous stress panel spanning different nuisance/discriminability regimes; it is not intended as an average benchmark over all control tasks.

Baselines. LeWM-base (no noise), LeWM+noise with full-sequence perturbed targets (8-level sweep), and a target-view ablation with perturbed input/history but original future targets (8-level sweep).

Checkpoints. The primary sweep contains 36 checkpoints: one no-noise baseline plus eight `std_max` values for each of the four tasks. The target-view ablation adds 32 nonzero-`std_max` checkpoints. The paired ACPC basin diagnostic is run on the 36 primary LeWM checkpoints.

Evaluation seeds. Each checkpoint is evaluated with 3 evaluation seeds (42 / 43 / 44), with 100 trajectories per seed.

Evaluation protocol. All LeWM rows use $n = 3$ eval seeds (42/43/44) and `num_eval = 100` trajectories per seed. Tables 1, 2, 3 report across-seed population mean ± std from `assets/paper1_data/canonical_evals_20260517.json`; Table 4 uses `assets/paper1_data/unperturbed_target_mainline_20260604.json`. Some artifacts use *clean* as the condition name; the claims use *unperturbed* or *original view*. Diagnostic source-of-truth for the primary sweep is `assets/paper1_data/canonical_diagnostics_20260517.json`.

Hardware. Single NVIDIA H800 (80 GB).

5.2 JEPA control fragility under visual corruptions

Table 1 reports LeWM-base success rates under unperturbed and noised evaluation (mean ± std across 3 seeds × 100 evaluations).

Table 1: LeWM-base under test-time visual corruptions (mean ± std; 3 seeds × 100 evaluations).

Task	unperturbed	pixels 0.05	pixels 0.08	px+goal 0.08	unperturbed → pixels 0.08 drop
TwoRoom	94.00 ± 3.56	72.33 ± 5.79	65.67 ± 1.25	50.00 ± 1.41	-28.33
PushT	86.33 ± 2.36	17.00 ± 1.41	4.33 ± 1.25	4.67 ± 2.05	-82.00
Reacher	58.67 ± 1.25	33.67 ± 2.87	18.33 ± 2.05	15.00 ± 2.16	-40.33
Cube	66.67 ± 2.62	48.67 ± 6.85	47.00 ± 6.38	46.33 ± 3.68	-19.67

LeWM-base is strong on unperturbed images (especially TwoRoom and PushT), but observation noise alone already produces large closed-loop drops: PushT loses 82.00 pts at pixels 0.08, Reacher 40.33 pts, TwoRoom 28.33 pts, and Cube 19.67 pts. The pixels+goal column shows the stronger full-visual-stream stress case; it further degrades TwoRoom and Reacher, but is not the primary robustness endpoint because the goal image is also perturbed. The drop pattern across tasks remains informative: Cube degrades least, while PushT degrades most catastrophically, confirming that contact-heavy continuous control is most sensitive to visual precision.

The same direction of failure is visible on PLDM but with task-dependent strength. PLDM trained without noise augmentation drops sharply on PushT (75.33% ± 3.68 unperturbed to 18.33% ± 2.05 at pixels 0.08, −57.00 pt) and TwoRoom (96.67% ± 1.70 to 67.00% ± 2.16, −29.67 pt), while Reacher and Cube have much smaller no-noise-training gaps (−2.33 and −4.33 pt; Table 16). The noise-training signatures still carry over: TwoRoom recovers into a high-noise plateau, PushT has a large `std_max` ≈ 0.03 recovery, and Cube responds weakly (Appendix F).

5.3 Noise augmentation closes the gap as a coarse pressure

Tables 2 and 3 report the complete 8-level sweep across tasks. All cells are success rate ± across-seed std (in pts), aggregated under the unified $n = 3$ seeds (42/43/44) × 100 trajectories protocol — 300 trajectories per cell.

Table 2: LeWM+noise sweep, unperturbed success rate (%; released condition-name: `clean`).

`std_max`	TwoRoom	PushT	Reacher	Cube
0 (base)	94.00 ± 3.56	86.33 ± 2.36	58.67 ± 1.25	66.67 ± 2.62
0.01	93.67 ± 3.30	88.00 ± 3.74	61.67 ± 2.49	69.33 ± 0.47
0.02	95.00 ± 2.83	88.33 ± 2.62	85.67 ± 2.49	60.00 ± 1.63
0.03	96.33 ± 3.30	89.67 ± 1.70	78.67 ± 1.25	65.00 ± 1.63
0.04	96.33 ± 2.05	89.33 ± 2.05	84.00 ± 2.94	69.00 ± 3.74
0.05	96.00 ± 2.83	80.67 ± 4.78	70.00 ± 2.16	59.33 ± 0.94
0.06	96.67 ± 2.05	89.33 ± 2.05	86.00 ± 2.94	66.67 ± 2.05
0.07	96.00 ± 1.63	85.67 ± 3.09	83.67 ± 3.30	67.67 ± 0.94
0.08	98.33 ± 0.47	88.33 ± 2.87	84.00 ± 0.82	62.33 ± 1.25

Table 3: LeWM+noise sweep, pixels 0.08 success rate with unperturbed goal (%).

`std_max`	TwoRoom	PushT	Reacher	Cube
0 (base)	65.67 ± 1.25	4.33 ± 1.25	18.33 ± 2.05	47.00 ± 6.38
0.01	94.00 ± 2.83	51.67 ± 5.44	45.67 ± 3.86	49.00 ± 2.45
0.02	94.33 ± 3.09	73.67 ± 4.99	83.67 ± 2.49	58.67 ± 3.30
0.03	97.00 ± 2.94	82.00 ± 6.48	78.67 ± 2.87	68.33 ± 2.49
0.04	96.67 ± 2.05	86.67 ± 2.87	82.00 ± 0.82	65.00 ± 2.45
0.05	95.33 ± 3.30	78.33 ± 5.91	70.00 ± 2.94	60.00 ± 0.82
0.06	96.33 ± 2.05	87.67 ± 3.30	81.33 ± 1.70	66.67 ± 1.70
0.07	96.67 ± 1.25	84.00 ± 3.74	84.67 ± 5.19	67.33 ± 2.05
0.08	97.67 ± 0.94	89.00 ± 4.32	82.00 ± 1.41	62.00 ± 1.41

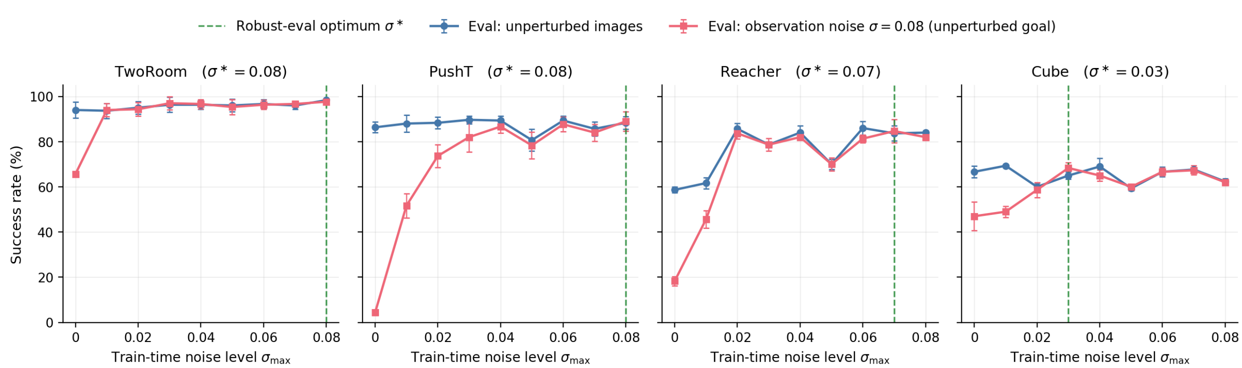

Figure 2: Noise-training sweep across all four tasks. Blue: success rate under unperturbed evaluation. Red: success rate under observation-only Gaussian noise $\sigma = 0.08$ with an unperturbed goal. Green dashed: each task's robust-eval optimum σ^* (the train-time `std_max` that maximises corrupted-eval success). The optimum σ^* varies across tasks, consistent with a coarse global scalar pressure with broad, task-dependent plateaus rather than a single jointly optimal level.

The sweep has two readings. First, input-side noise is effective: PushT recovers from 4.33% to 89.00% at pixels 0.08, and Reacher from 18.33% to 84.67%. Second, the pressure is coarse and task-dependent. TwoRoom favours heavy noise, PushT's unperturbed and noisy-eval optima dissociate (0.03 vs 0.08), Reacher has a broad plateau, and Cube responds weakly. Several optima are plateau regions rather than isolated points, so the claim is not that exact per-task tuning is always necessary; it is that one scalar perturbation strength cannot express the selective-consistency tradeoff by itself.

5.4 Target-view ablation: original future targets are not enough

The ACPC framing suggests a simple first training ablation: keep the input/history perturbed but encode the prediction target from the original future frame. This branch changes only the prediction-target view; it does not add a paired rollout-consistency term or a discriminability guard. We trained this perturbed-input → original-target branch for all four tasks and all nonzero `std_max` values, yielding 32 additional LeWM checkpoints. Every row has the same three-seed evaluation protocol and a completed diagnostics summary; the released aggregate is `assets/paper1_data/unperturbed_target_mainline_20260604.json`.

Table 4: Target-view ablation on LeWM. Each entry is the checkpoint that maximises pixels 0.08 success within the branch. Full-sequence perturbed targets remain much stronger on the primary robust endpoint than the perturbed-input → original-target branch.

Task	Full-sequence perturbed target			Perturbed input → original target		
	best `std_max`	origin	pixels 0.08	best `std_max`	origin	pixels 0.08
TwoRoom	0.08	98.33 ± 0.47	97.67 ± 0.94	0.04	93.00 ± 2.94	73.67 ± 3.40
PushT	0.08	88.33 ± 2.87	89.00 ± 4.32	0.02	88.33 ± 2.05	8.33 ± 1.70
Reacher	0.07	83.67 ± 3.30	84.67 ± 5.19	0.07	59.67 ± 5.31	30.00 ± 3.56
Cube	0.03	65.00 ± 1.63	68.33 ± 2.49	0.02	66.67 ± 1.25	42.67 ± 5.44

This is a negative ablation, and it matters for the story. The original-target branch slightly improves over no-perturbation training on TwoRoom, PushT, and Reacher, but it does not approach the full-sequence branch: the pixels 0.08 gap is 24.00 points on TwoRoom, 80.67 on PushT, 54.67 on Reacher, and 25.67 on Cube. Thus the natural denoising-style target policy is not a sufficient implementation of ACPC. The result keeps full-sequence perturbation as the stronger empirical baseline while narrowing the method implication: future training should regularise paired action-conditioned predictions directly, with a discriminability guard, instead of assuming that changing only the prediction-target view will make nuisance perturbations collapse in the right place.

The behavioural evidence for this selective-consistency tension is summarised by the sweep tables, Figure 2, and the target-view ablation in Table 4. Before turning to the broader five-layer diagnostic suite, we first report a paired Gaussian-noise ACPC basin diagnostic that directly checks whether the same-state noisy views occupy a smaller action-conditioned predictive region after noise training.

5.5 Paired Gaussian-noise ACPC basin diagnostic

We compute an eval-only ACPC basin diagnostic on the same 36 LeWM epoch-10 checkpoints used in the sweep. For each checkpoint we sample 100 fixed dataset windows and create eight paired same-state views using Gaussian pixel noise with std $\in \{0.01, \ldots, 0.08\}$, matching the training sweep's corruption family. Each original/noised branch is encoded and then rolled out under the same recorded action sequence for horizon $H = 8$. The released source is `assets/paper1_data/acpc_basin_diagnostics.json`, generated by `tools/paper1_acpc_basin.py`. The diagnostic intentionally excludes blur/resize so that the ACPC evidence is not confounded by train/eval corruption-family mismatch.

For each checkpoint, let R_E be the median pairwise spread among the original and noised encoder views, normalised by the original-view local NN L2 scale, and let R_F be the median pairwise spread among their predicted rollouts, normalised by the original transition L2 scale. Lower R_F means the same-state perturbation views induce a tighter action-conditioned predictive basin; R_F/R_E is a descriptive contraction ratio rather than a success predictor.

Table 5: Gaussian-noise ACPC basin diagnostic on LeWM: no-noise baseline vs. each task's pixels 0.08 robust-best checkpoint. R_E is normalised encoder-view spread; R_F is normalised action-conditioned rollout-view spread under the same action sequence. The diagnostic uses Gaussian-noise views of the observation history (std 0.01–0.08) while leaving the goal unperturbed, matching the primary unperturbed-goal evaluation.

Task	ckpt	pixels 0.08	drop	R_E	R_F
TwoRoom	base	65.67	28.33	3.538	0.785
TwoRoom	noise 0.08	97.67	0.67	0.235	0.028
PushT	base	4.33	82.00	1.727	1.543
PushT	noise 0.08	89.00	−0.67	0.137	0.088
Reacher	base	18.33	40.33	4.238	1.012
Reacher	noise 0.07	84.67	−1.00	0.135	0.027
Cube	base	47.00	19.67	1.343	0.957
Cube	noise 0.03	68.33	−3.33	0.238	0.115

The strict reading is important. The diagnostic supports the ACPC framing because robust noise-trained checkpoints have much smaller action-conditioned prediction basins under same-family visual perturbations: e.g., PushT reduces R_F from 1.543 to 0.088, and TwoRoom from 0.785 to 0.028. It does *not* support the stronger universal claim that encoder representations remain widely separated while predictions alone collapse them. In this Gaussian-noise setting, robust checkpoints usually shrink both the encoder basin and the predictive basin, with the predictor-space radius remaining the control-facing quantity. This is why we use the basin diagnostic to refine the mechanism claim, not to replace the behavioural corruption evaluation.

The broader representation-level evidence appears next in Table 6.

5.6 Diagnostic analysis: why global noise is only a coarse pressure

Where the per-task sweep evidence in Section 5.3 is the behavioural face of selective consistency, the diagnostics in this subsection are its representational face: which latent properties move under noise training, and whether the movement is bounded by each task's discriminability requirements. Table 6 compares core diagnostic metrics on LeWM-base versus the per-task representative noise-trained diagnostic checkpoint.

Table 6: Representation diagnostics: LeWM-base vs. a representative noise-trained diagnostic checkpoint per task. The σ pick per task (0.08 / 0.02 / 0.06 / 0.01) is the ckpt on which the diagnostic suite was originally executed. Under the unified 3-seed $\times$ 100 eval protocol the PushT unperturbed point-best shifts to `std_max` = 0.03 (within ± 2 pt of `std_max` = 0.02); we keep the diagnostics on the `std_max` = 0.02 checkpoint because the compression-vs.-resolution pattern this table illustrates is robust within this neighbourhood.

Metric	TwoRoom base	TwoRoom noise (0.08)	PushT base	PushT noise (0.02)	Reacher base	Reacher noise (0.06)	Cube base	Cube noise (0.01)
`clean_nn_cos_dist_median`	0.0449	0.0281	0.2360	0.1051	0.0633	0.0676	0.1856	0.1879
`clean_effective_rank`	47.60	33.59	76.42	42.85	61.04	65.92	73.25	71.83
`transition_resolution_ratio_l2`	0.7216	0.6055	0.3015	0.2800	0.3704	0.3791	0.4847	0.4629
`transition_resolution_ratio_cos`	0.5538	0.3780	0.0868	0.0800	0.1351	0.1399	0.2347	0.2168
`id_probe_r2`	0.2889	-0.0573	0.7739	0.7500	0.1621	0.1729	0.6657	0.6720
`action_mean_pred_shift_norm`	0.5329	0.4482	0.1283	0.1200	0.2518	0.2585	0.2364	0.2320
`predictor_rollout_T8_l2`	18.62	17.90	18.65	16.50	15.17	0.44	20.20	19.25

Notes. (i) Reacher and Cube representative diagnostics are taken from the corresponding checkpoint's per-tool JSON outputs under the diagnostics directory (max-std = 0.1, history-only noise); (ii) the Reacher representative rollout drift of 0.44 (a 35$\times$ reduction from base 15.17) is not a recording error: Reacher's representative checkpoint trains under `std_max` = 0.06, which is enough to collapse the multi-step predictor drift while keeping single-step task-resolution metrics flat — precisely the dissociation that motivates the analysis in Section 5.7. Cube's representative drift (19.25 $\approx$ base 20.20) is the opposite extreme.

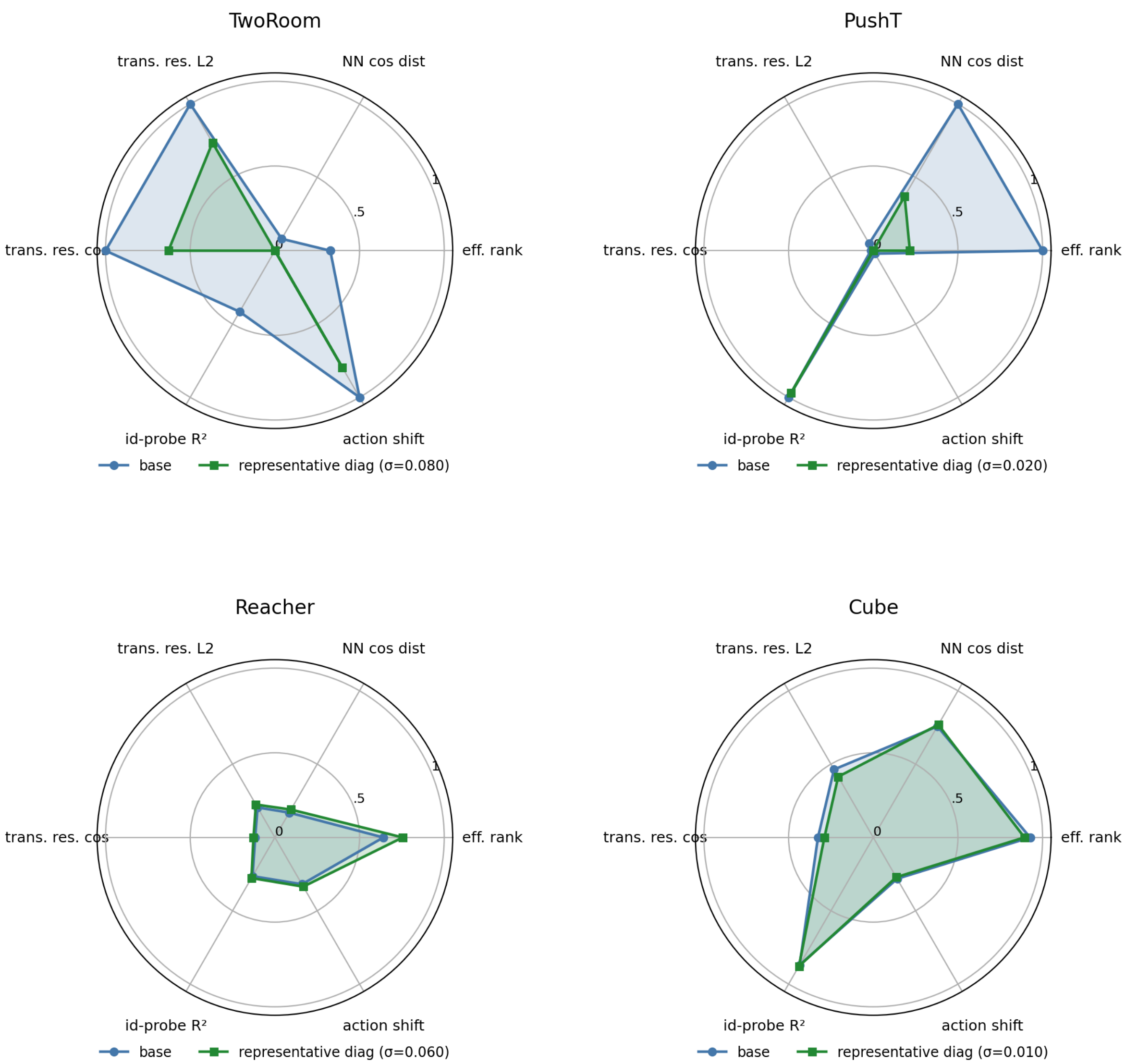

Figure 3: Per-task diagnostic radar: base vs representative noise-trained diagnostic checkpoint on 6 metrics, min-max normalised across all 4 tasks.

Mechanistic reading.

- **TwoRoom.** Low-dimensional, discrete, visually redundant. Compressing the representation (effective rank $47.6 \to 33.6$) is acceptable and even beneficial: a more compact latent space is easier to plan in.
- **PushT.** Continuous contact requires fine-grained pose resolution. Even at the representative diagnostic checkpoint (`std_max` $= 0.02$), the L2 transition-resolution metric already trends slightly downward. At heavier noise (e.g. 0.06) it drops further, erasing contact-transition keyframes.
- **Reacher.** Low-dimensional continuous reaching does not show a resolution collapse in Table 6: effective rank, transition resolution, and the controllability probe remain flat or slightly improve. The notable change is the large reduction in multi-step predictor drift ($15.17 \to 0.44$), matching the later finding that Reacher's residual corruption-drop signal lives in the rollout metric rather than in the single-step fragility ratio.
- **Cube.** Cube is moderately resolution-demanding but less fragile than PushT. The representative

checkpoint leaves rank, transition resolution, controllability probe, and rollout drift almost unchanged, consistent with Cube's smaller visual-corruption cliff and the moderate residual signal of multi-step drift in Table 8.

- **Predictor-rollout caveat.** A drop in multi-step predictor rollout drift is not unambiguously good news. It can also mean the latent has become more *predictable* without being more *controllable*: predictor stability bought by sacrificing resolution.

5.7 Cross-checkpoint correlation analysis

Table 7 reports task-specific cross-checkpoint correlations on the LeWM $n = 9$ sweep using the unified 3-seed $\times$ 100 evaluation.

We analyse two single-value-per-checkpoint diagnostic metrics that have full coverage on all 9 checkpoints per task: the fragility ratio, defined in Section 4.3 as the single-step predictor target shift normalised by the nearest-neighbour cosine distance at the diagnostic's max-std injection, and the corresponding multi-step rollout L2 drift. Both are released in `assets/paper1_data/canonical_diagnostics_20260517.json`, derived from each checkpoint's predictor-sensitivity diagnostic JSON, and are pure checkpoint-level quantities (independent of the evaluation protocol).

Table 7: LeWM $n = 9$ sweep: per-task Pearson r / Spearman ρ vs corruption drop (unperturbed − pixels 0.08, unperturbed goal). Eval values come from the unified 3-seed $\times$ 100 protocol.

Metric	TwoRoom (r/ρ)	PushT (r/ρ)	Reacher (r/ρ)	Cube (r/ρ)
`predictor_target_to_nn_cos_ratio_at_max_std`	+0.92/+0.20	−0.55/−0.80	+0.98/+0.55	+0.39/+0.27
`predictor_rollout_T8_l2_at_max_std`	+0.81/+0.31	+0.98/+0.93	+0.99/+0.58	+0.91/+0.48

Reading. Unconditional correlations are large because both diagnostic metrics and the corruption drop are driven by the same upstream variable — `std_max`. The relevant test is partial correlation conditioned on `std_max` (Table 8).

Table 8: Partial Spearman ρ(metric, corruption drop | `std_max`), $n = 9$ per task.

Metric	TwoRoom	PushT	Reacher	Cube
`predictor_target_to_nn_cos_ratio_at_max_std`	−0.03	+0.19	+0.27	+0.12
`predictor_rollout_T8_l2_at_max_std`	0.00	−0.01	+0.37	+0.23

After removing the monotone sweep trend associated with `std_max`, the **fragility metric carries little stable information about the size of the corruption drop** on any task; all bootstrap intervals for the partial correlations include zero. The **multi-step predictor drift** retains only weak residual signal under the unperturbed-goal pixels 0.08 endpoint (Reacher +0.37, Cube +0.23, both with wide intervals). This weakens the older pixels+goal-stress reading and is the reason we treat the cross-checkpoint diagnostic as mechanism localisation, not as a robustness predictor.

Table 9: Spearman ρ (unconditional) and partial Spearman ρ conditioned on `std_max`, for the fragility ratio on the PushT $n = 9$ LeWM sweep under the unified 3-seed $\times$ 100 eval. CIs are the percentile 95% interval over $B = 1000$ bootstrap resamples (with replacement, seed = 42; see Section 4.4). The `std_max`-only top four rows are univariate sweep diagnostics with no metric/outcome pairing, so no CI is reported.

Quantity	Spearman ρ	95% bootstrap CI
ρ(`std_max`, metric)	+0.83	—
ρ(`std_max`, unperturbed)	0.00	—
ρ(`std_max`, pixels 0.08)	+0.88	—
ρ(`std_max`, corruption drop)	−0.98	—
ρ(metric, unperturbed) unconditional	−0.33	$[-0.95, +0.62]$
ρ(metric, unperturbed) | `std_max` partial	−0.59	$[-0.97, -0.10]$
ρ(metric, pixels 0.08) unconditional	+0.60	$[-0.11, +1.00]$
ρ(metric, pixels 0.08) | `std_max` partial	−0.53	$[-0.84, +0.00]$
ρ(metric, corruption drop) unconditional	−0.80	$[-1.00, -0.23]$
ρ(metric, corruption drop) | `std_max` partial	+0.19	$[-0.00, +0.70]$

PushT $n = 9$ detailed view (fragility ratio). Interpretation:

1. **After removing the monotone trend associated with `std_max`, the metric still carries a residual checkpoint-quality signal.** Partial ρ on unperturbed evaluation is -0.59 and on pixels 0.08 is -0.53 — both negative. On the $n = 9$ PushT sweep, lower fragility ratio aligns with better unperturbed *and* better noisy success beyond what the sweep-level `std_max` trend already explains.
2. **The metric does NOT predict the unperturbed-to-corrupted gap.** The unconditional $\rho(\text{metric}, \text{drop}) = -0.80$ looks impressive, but partialling out `std_max` reduces it to $+0.19$ (95% bootstrap CI $[-0.00, +0.70]$, including 0). The drop's strong correlation with the metric is a mediated effect: `std_max` drives both the metric ($\rho = +0.83$) and the drop ($\rho = -0.98$). After the `std_max` trend is removed, the metric tells you little stable information about how much the *gap* between unperturbed and noisy performance will be.
3. **Note the unconditional-vs-partial sign reversal on unperturbed evaluation.** $\rho(\text{metric}, \text{unperturbed})$ is only -0.33 unconditionally — unperturbed success rate is roughly flat across the PushT sweep under the 3-seed protocol (range 80.67–89.67), so `std_max` barely moves unperturbed performance. The partial correlation *strengthens* to -0.59 once the sweep-level `std_max` trend is removed, exactly because the residual signal is what the metric tracks.
4. **Practical reading.** The toolkit is useful for mechanism localisation and checkpoint selection after controlling for the sweep-level `std_max` effect. It is *not* a substitute for actually training and evaluating under corruptions when the robustness gap is the quantity of interest.

5.7.1 Unperturbed control fidelity and corruption robustness as separable signals

The partial-correlation analysis above settles the interpretation:

- The metric is a **residual checkpoint-quality signal beyond the sweep-level `std_max` effect** — a lower fragility ratio means better control on *both* unperturbed and noisy evaluation (partial $\rho = -0.59$ / -0.53 on PushT).
- The metric **does not isolate noise robustness** — its apparent correlation with the gap between unperturbed and noisy success is mediated by `std_max` (partial $\rho = +0.19$, wide CI including zero).

The PushT scatter (Figure 4) makes both halves visible. Panel (a) plots metric vs unperturbed success (unconditional Spearman $\rho = -0.33$; the residual trend after conditioning on `std_max` is what the partial correlation captures). Panel (b) plots metric vs corruption drop (unconditional $\rho = -0.80$), but the colour-bar (encoding `std_max`) reveals the structure: low-`std_max` checkpoints (light blue) live at high drop, high-`std_max` checkpoints (dark blue) live at low drop, and the metric tracks `std_max` itself. After the `std_max` trend is removed, the metric does not robustly separate small-drop from large-drop checkpoints.

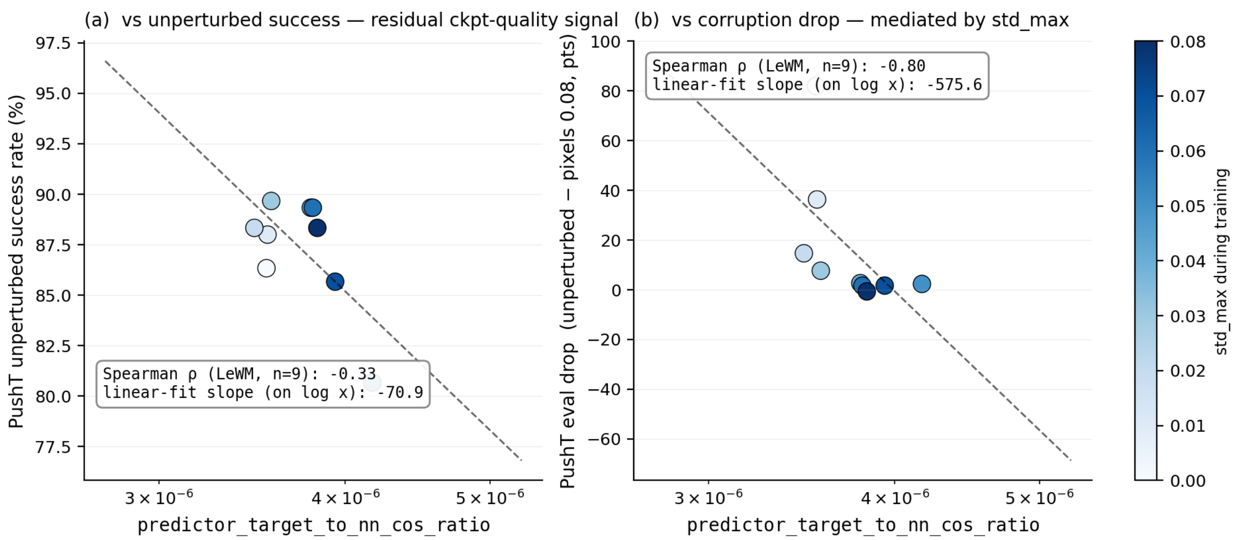

Figure 4: PushT $n = 9$ LeWM sweep: the fragility ratio is a checkpoint-quality predictor (a); the apparent corruption-drop correlation in (b) is mediated by `std_max` (encoded by the colour bar).

5.8 Mechanism attribution: where in the pipeline does noise cause failure?

Section 5.6 reports *what* is compressed and Section 5.7 reports *which diagnostics* correlate with eval across checkpoints, but neither answers **"is the failure in the encoder, the predictor, or the cost surface?"** We address this with two complementary experiments.

5.8.1 Auxiliary cost-swap sanity check: cost alone is unlikely to explain the collapse

We ran a one-off eval-only cost swap on a single TwoRoom checkpoint outside the 36-checkpoint canonical evaluation table; full details are reported in Appendix E. Switching the CEM cost from cosine/normalized to mse/raw changes px+goal 0.03 success only from 36.0 to 42.0, still far below a separate unperturbed reference of 69.7 (separate $n = 300$ eval, single seed; see Appendix E). As a sanity check, this suggests that the cost function alone is unlikely to explain the visual-corruption collapse.

5.8.2 Latent-noise probing: encoder shift is a major contributor

Directly injecting noise into the latent z (skipping the encoder) decouples encoder contributions from predictor + cost. Comparing diagnostic signals across two injection points (pixels vs. latent z, both history-only noise at max std):

- **PushT.** Multi-step input-space rollout drift has unconditional $\rho = +0.93$ vs corruption drop (Table 7), driven primarily by `std_max`; after removing the monotone `std_max` trend, the partial ρ collapses to -0.01. The single-step fragility ratio (unconditional $\rho = -0.80$; partial $+0.19$ after removing the same trend) tells the same story. Both encoder–predictor signals are mediated by training noise; once that sweep-level effect is accounted for, neither isolates the corruption drop on PushT.

- **Reacher.** Multi-step rollout drift has unconditional $\rho = +0.58$ and partial $\rho = +0.37$ vs corruption drop (95% bootstrap CI $[-0.35, +0.99]$). Under the unperturbed-goal observation-noise endpoint, this is only a weak residual signal.
- **Cube.** Multi-step rollout drift has unconditional $\rho = +0.48$ and partial $\rho = +0.23$ (95% CI $[-0.44, +0.96]$), so Cube's residual is also weak.
- **TwoRoom.** Partial correlations are finite but near zero after conditioning on `std_max`; saturation still limits what can be inferred from cross-checkpoint ranks.

Table 10 summarises the three-layer attribution.

Table 10: Three-layer attribution by task (LeWM $n = 9$ sweep, unified 3-seed $\times$ 100 eval).

Task	Strongest signal in these probes	Residual after removing the `std_max` trend
TwoRoom	encoder-shift signal (rank-saturated)	n/a
PushT	encoder + single-step predictor (both mediated by `std_max`)	no residual metric isolates corruption drop
Reacher	encoder + multi-step rollout	weak multi-step drift residual signal (partial $\rho = +0.37$)
Cube	encoder	weak residual on multi-step drift (partial $\rho = +0.23$)

Across these probes, the strongest common signal is **encoder shift transduced by the predictor**. The auxiliary cost-swap sanity check in Section 5.8.1 suggests that the cost function alone is unlikely to explain the collapse, but we do not treat that single-checkpoint ablation as a task-wide quantitative attribution. A stronger causal claim would require multi-task latent-injection or cost-ranking ablations. The present evidence is sufficient for mechanism localisation: once the encoder's latent-neighbourhood structure is corrupted beyond the NN distance scale, downstream predictor and planner can operate on the wrong neighbourhood.

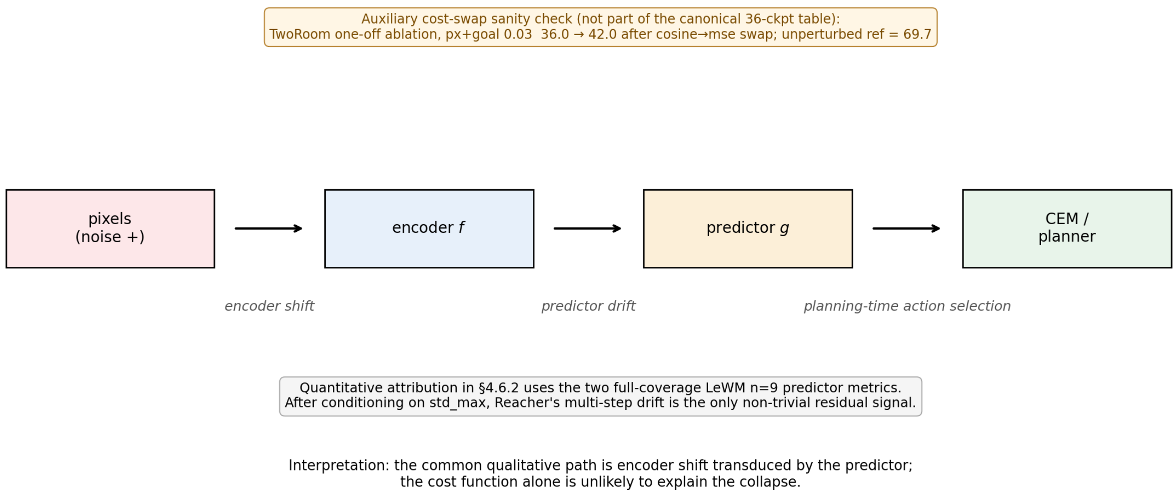

Figure 5: Mechanism schematic. Pixels perturb the encoder, the predictor transduces that shift into planning-relevant latent drift, and CEM acts on the resulting latent geometry. Quantitative attribution in §5.8.2 uses the two full-coverage LeWM $n = 9$ predictor metrics; after conditioning on `std_max`, Reacher's multi-step drift is the only non-trivial residual signal.

6 Discussion

6.1 From latent invariance to predictive consistency

The data challenge an implicit community assumption: latent prediction alone does not confer robustness to visual noise. But the corrective is *not* to demand that corrupted and unperturbed images map to the same latent. Our results are better read through the predictive-consistency lens of Section 3: encoder-level latent closeness is neither necessary nor sufficient (Section 3.1), so the target is not $z \approx \tilde{z}$ but agreement of the action-conditioned predictions, $F^k(z, \mathbf{a}) \approx F^k(\tilde{z}, \mathbf{a})$, in task-relevant coordinates. The invariance a JEPA encoder acquires is *in-distribution* invariance; when test-time inputs carry pixel noise unseen during training, the latent neighbourhood structure breaks down, the predictor rolls out the wrong future, and the planner fails. What matters for control is whether same-state perturbations stay predictive-consistent while action-sensitive state differences stay discriminable.

This should be read alongside VJEPA [16], which reports that JEPA-family models keep signal-recovery $R^2 > 0.84$ under a synthetic Noisy-TV distractor. That positive result is compatible with ours because the evaluation modalities differ: signal-recovery probes only need the latent to preserve recoverable state information, whereas closed-loop control demands a representation *and* predictor that support precise action optimisation — exactly the action-conditioned predictions on which we place the consistency requirement.

6.2 Task-dependent manifestation and scope

The selective-consistency demand — contract nuisance perturbations of the same state, preserve action-relevant transitions — has the following four-task signature in this paper. We use two qualitative task-property labels below: "nuisance-contraction demand" is the room to contract irrelevant visual variation, and "control-discriminability demand" is the action-relevant information that must survive.

- **TwoRoom.** Background wall colour / texture is pure visual redundancy; discarding it does not impair navigation. This does not mean that all positions can collapse: room, doorway, and goal-topology distinctions must remain separable.
- **PushT.** The pixel changes during T-block / arm contact carry critical force and pose information; discarding them blinds the planner.
- **Reacher.** Low-dimensional continuous control; visual encoding of joint angle requires moderate resolution.
- **Cube.** Object pose and grasp-point spatial relations need moderate resolution, but Cube itself is largely insensitive to global noise (action sequence is structured; visual–action coupling is predictable).

Task-specificity is why input-side noise behaves as a coarse global pressure rather than a single jointly optimal level. Table 11 consolidates the behavioural (Section 5.3) and representational (Section 5.6) evidence into a single per-task read.

Table 11: Where each LeWM task sits under the selective-consistency tension. The two "demand" columns are qualitative task-property reads, not independent measurements; they index the underlying behavioural and representational evidence in Sections 5.3 and 5.6 and the operational definition in Section 4.3. Sweep signatures come from Tables 2 and 3; diagnostic readings from Table 6.

Task	Nuisance-contraction demand	Control-discriminability demand	Observed sweep signature	Diagnostic reading at representative ckpt
TwoRoom	high (visual redundancy)	low–moderate (discrete navigation)	heavy-noise plateau ($\sigma^* = 0.08$)	compact latent acceptable; rank 47.6 → 33.6 without controllability loss
PushT	moderate	high (contact precision)	dissociated optima ($\sigma^*_{\text{unp}} = 0.03$, $\sigma^*_{\text{corr}} = 0.08$)	compression risks resolution; rank 76.4 → 42.9 with transition-resolution and inverse-dynamics-probe downward movement
Reacher	moderate	moderate	broad plateau ($\sigma^* \in 0.02$–0.07)	rollout drift carries weak residual signal more than rank or single-step resolution
Cube	low–moderate	moderate	weak response; pixels 0.08 point-best at $\sigma^*_{\text{corr}} = 0.03$	structured action coupling buffers noise; small representation movement

Scope. The task-level fragility/recovery signature is replicated on PLDM, but the mechanism claim is LeWM-centred. Reconstruction-based world models, decoder-free latent MPC systems, EMA-target JEPAs, and variational JEPAs may exhibit different compression dynamics.

6.3 When the diagnostic toolkit works — and when it does not

Three conditions bound the toolkit. First, it can rank residual checkpoint quality after conditioning on `std_max`, but it does not isolate corruption-specific robustness: on PushT, fragility ratio retains signal for unperturbed and noisy success yet not for the unperturbed-to-corrupted gap. Second, tasks with little residual checkpoint spread after the sweep trend, such as Reacher and TwoRoom here, fall outside the strict diagnostic regime. Third, no static diagnostic can replace the training intervention itself; whether the model saw perturbations during training dominates the corruption drop. The toolkit is therefore an unperturbed-evaluation auxiliary for mechanism localisation and checkpoint selection within a fixed protocol, not a robustness oracle.

6.4 Practical guidance: how to choose `std_max` on a new task

Use the diagnostics as a screening tool, then run direct eval at both unperturbed and high-corruption endpoints. Check fragility ratio, effective rank, transition resolution, and task semantics together; none is a robustness oracle alone. A practical search is a coarse sweep such as {0.01, 0.03, 0.05, 0.07}, followed by refinement around the best unperturbed/corrupted candidates.

6.5 Implications for target-view and predictive-dynamics objectives

The measured target-view ablation in Table 4 rules out a tempting shortcut: perturbed-history $\to$ original-future prediction is not enough to recover robust control. This does not invalidate the ACPC definition, because the branch changes only the target view of the one-step prediction loss; it does not explicitly align paired action-conditioned rollouts or protect action-relevant distinctions. The next method objective should therefore regularise paired predictive dynamics after the action-conditioned predictor and gate that pressure by action/transition sensitivity. The negative heteroscedastic result below is the warning: high prediction error can mark action-critical contact transitions, so prediction difficulty alone is the wrong gate.

6.6 Limitations and future directions

Limitation 1 — architecture scope. The main diagnostics are LeWM-centred; PLDM replicates task-level signatures and the PushT partial-correlation null, but its internal recovery mechanism differs. EMA-target and variational JEPA variants remain open.

Limitation 2 — corruption scope. Gaussian pixel noise is the controlled training axis. Blur stress tests in Appendix G show visual fragility beyond Gaussian noise, but also corruption-specific task ordering; blur-specific training remains follow-up work.

Limitation 3 — diagnostic scope. The framework is empirical. Reacher and TwoRoom fail the partial-correlation criterion for all metrics, and we do not yet have a theory predicting which tasks support cross-checkpoint diagnostics.

Limitation 4 — method scope. The main paired ACPC basin diagnostic is LeWM/Gaussian-noise only. Appendix H broadens the probes under auxiliary pixels+goal stress, but those rows are post-hoc, small-sample, and exploratory. The target-view ablation is now measured and negative; a method paper still needs a trained paired predictive-dynamics consistency objective.

Additional ablation — hard $\neq$ nuisance. A heteroscedastic NLL probe with a learned per-transition σ-head confirms that error-based downweighting is unsafe: TwoRoom remains strong, but PushT unperturbed success collapses from 86.33% to 13.33% because contact-control transitions are high-error and action-critical. Full data are in Appendix D.

Future direction 1 — paired predictive-dynamics training. Since the original-target branch underperforms full-sequence perturbation, the next objective is not another target-view-only loss. It is paired ACPC_H regularisation after the action-conditioned predictor, gated by action sensitivity and protected by a discriminability guard.

Future direction 2 — broader replication. Frozen/pretrained visual-feature models, EMA-target JEPAs, and non-Gaussian training corruptions remain the next checks.

Future direction 3 — Theoretical grounding. Reformulating the consistency/discriminability tension in an information-bottleneck / rate-distortion language is an attractive direction; we did not pursue it because the empirical phenomenology may not yet be stable enough for formal modelling.

7 Conclusion

This paper reframes visual robustness for JEPA world-model control as action-conditioned predictive consistency with an explicit discriminability countercondition. Controlled Gaussian corruptions show that latent prediction alone does not confer robustness; input-side noise augmentation recovers much of the lost control performance, but only as a coarse pressure with task-dependent plateaus. The paired ACPC basin diagnostic gives the control-facing reading: robust checkpoints induce tighter same-action prediction basins under same-state perturbations, while pointwise fragility does not isolate the corruption gap after conditioning on train-time noise.

We do not propose a new training algorithm. The evidence now rules out a naive target-view-only implementation: perturbed-history → original-future targets underperform full-sequence perturbed targets on all four robust endpoints. The staged method path is therefore narrower and clearer: train paired predictive-dynamics consistency after the action-conditioned predictor, gated by action sensitivity and protected by discriminability.

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The complete code and data are available at `https://github.com/qun-team/wm_exp`. We thank the LeWM authors for releasing the baseline framework on which this study builds.

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A Experimental details

A.1 Environments

- **PushT.** 2D continuous pushing; 20,000 expert episodes; $\sim$196 steps/episode; action dim 2 (direction + magnitude); 224×224 RGB.
- **TwoRoom.** 2D continuous navigation; 10,000 episodes; $\sim$92 steps; action dim 2; 224×224 RGB.
- **Reacher.** 2D arm control (DeepMind Control Suite); 10,000 episodes; 200 steps; action dim 2.
- **Cube.** 3D cube manipulation (OGBench); 10,000 episodes; 200 steps; action dim 7.

A.2 Noise augmentation implementation

The actual implementation is `utils.py::AddNormalizedGaussianNoise`. Key points: (i) noise is added on ImageNet-normalized tensors and must be divided by the per-channel std to retain "pixel-space std" semantics; (ii) sampling is per-frame independent over the leading frame dims, not per-batch; (iii) all sweeps in this paper fix `noise_prob` = 1.0 and `std_min` = 0 and vary only `std_max`.

```
class AddNormalizedGaussianNoise:
    """Per-frame independent. Each frame draws Bernoulli(noise_prob) then
    std ~ Uniform(std_min, std_max). Pixel-space std is converted to
    normalized space by dividing by per-channel std before adding."""
    def __init__(self, std_min, std_max, noise_prob=1.0):
        self.std_low, self.std_high, self.noise_prob = std_min, std_max, noise_prob
        self.channel_std = torch.as_tensor(IMAGENET_STD) # (C,)

    def __call__(self, x):
        if self.std_high <= 0 or self.noise_prob <= 0:
            return x
        leading = x.shape[:-3]
        stds = torch.empty(leading, device=x.device, dtype=x.dtype).uniform_(
            self.std_low, self.std_high)
        if self.noise_prob < 1.0:
            mask = (torch.rand(leading, device=x.device) < self.noise_prob).to(x.dtype)
            stds = stds * mask
        per_frame_scale = stds.view(*leading, 1, 1, 1)
        channel_factor = (1.0 / self.channel_std.to(x.device, x.dtype)).view(
            *([1] * len(leading)), -1, 1, 1)
        scale = per_frame_scale * channel_factor # pixel-space std -> normalized
        return x + torch.randn_like(x) * scale
```

A.3 Evaluation protocol

Unperturbed evaluation uses no noise, 100 trajectories per seed $\times$ 3 seeds (42/43/44), totalling 300 trajectories per condition. The primary noisy evaluation adds Gaussian noise of std $\in \{0.05, 0.08\}$ to observation pixels while leaving the goal image unperturbed under the same 3-seed protocol. Pixels+goal corruption is retained as an auxiliary full-visual-stream stress condition and is reported separately where used. Success criterion is task-specific (PushT: T-block pose match; TwoRoom: target region; Reacher: joint-angle match; Cube: cube position match).

A.4 Compute

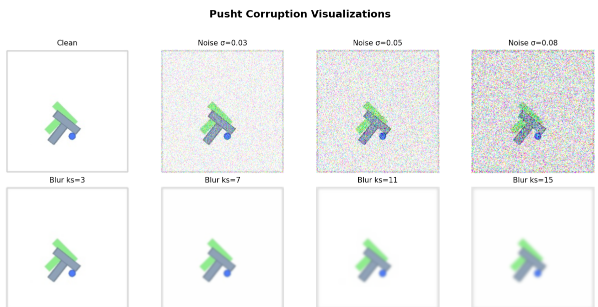
Training runs on a single NVIDIA H800 (80 GB) and follows the LeWM training schedule released with the baseline.

A.5 Main-figure rendering

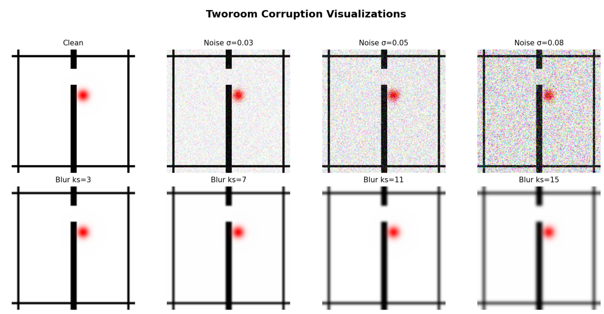
Figures 2 to 5 are produced by `tools/paper1_figs.py`; run `python-mtools.paper1_figs--out-dirassets/paper1_figs`. The script reads `assets/paper1_data/canonical_evals_20260517.json` together with `assets/paper1_data/canonical_diagnostics_20260517.json`; no checkpoint or model loading is required.

A.6 Corruption visualizations

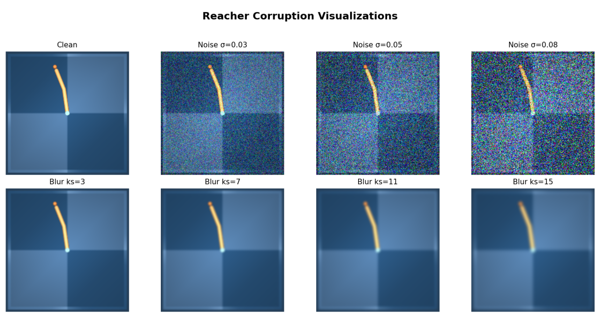
Figure 6 shows representative renderings of the two evaluated corruption families, applied to a sample observation from each of the four tasks. The Gaussian-noise row includes a mild calibration severity $\sigma = 0.03$ in addition to the two evaluation endpoints $\sigma \in \{0.05, 0.08\}$ used in Sections 5.2 and 5.3; the main sweep treats observation-only noise with an unperturbed goal as the primary endpoint and keeps pixels+goal noise as an auxiliary stress condition. The Gaussian-blur kernel sizes ($k \in \{3, 7, 11, 15\}$) match the stress test in Appendix G. Each panel is a single 224×224 RGB frame from the corresponding task's evaluation rollout, after corruption and before encoder normalisation.



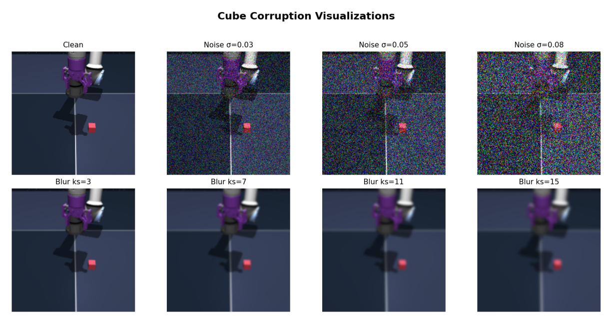
(a) PushT



(b) TwoRoom



(c) Reacher



(d) Cube

Figure 6: Corruption visualizations on the four tasks. Each task panel is a 2×4 grid: **Row 1** additive Gaussian pixel noise (the *Original* reference and $\sigma \in \{0.03, 0.05, 0.08\}$); **Row 2** Gaussian blur at kernel sizes $k \in \{3, 7, 11, 15\}$ (sigma derived via the OpenCV / torchvision auto rule). $\sigma = 0.08$ is a strong stress-test endpoint at the 224×224 native input resolution; Table 1 reports the no-noise-training control success rate at this endpoint, and Tables 2 and 3 report the recovery achievable under noise-augmented training.

B Full diagnostic profile of LeWM-base on four tasks

Table 12 summarises every core diagnostic on the four LeWM-base checkpoints. Data are taken from each checkpoint's per-tool JSON outputs under `eval_results/diagnostics/` (the `geometry`, `task`, `predictor` and `latent_noise` families).

Table 12: Full LeWM-base diagnostics across four tasks.

Layer / Metric	TwoRoom	PushT	Reacher	Cube	Unit / note
Encoder geometry					
`clean_nn_cos_dist_median`	0.0449	0.2360	0.0633	0.1856	cosine distance
`clean_pair_cos_dist_median`	0.9904	1.0228	1.0252	1.0193	pair-wise cos dist
`clean_effective_rank`	47.60	76.42	61.04	73.25	effective rank
Noise sensitivity					
`noise_angle_deg_median` (@std= 0.05)	5.51°	1.33°	3.22°	1.40°	median angular shift
`noise_to_nn_cos_ratio_median`	0.1031	0.011	0.0249	0.016	noise/NN cos ratio
`robust_radius_std`	0.0142	0.0537	0.0142	0.0356	critical noise std
`first_risk_std`	>0.08	>0.08	>0.08	>0.08	first high-risk std
`noise_angle_slope_deg_per_std`	1085.8	284.8	831.7	327.0	°/std angular gain
`geometry_flag`	balanced	robust	balanced	robust	geometry label
Task resolution					
`transition_resolution_ratio`_l2	0.7216	0.3015	0.3704	0.4847	L2 resolution ratio
`transition_resolution_ratio`_cos	0.5538	0.0868	0.1351	0.2347	cos resolution ratio
`id_probe_r2`	0.2889	0.7739	0.1621	0.6657	linear probe R^2
`id_probe_r2_min`	0.2599	0.6786	0.1366	0.0972	min probe R^2
`lidar_rank`	46.06	13.95	45.90	42.46	LiDAR rank proxy
Action effect					
`action_mean_pred_shift_norm`	0.5329	0.1283	0.2518	0.2364	action-perturb mean shift
`action_perturb_pred_shift_corr`	0.2847	0.2873	0.4042	0.2559	shift $\times$ $\lVert a \rVert$ corr
Predictor stability					
`predictor_rollout_T8_l2`	18.62	18.65	15.17	20.20	T=8 rollout L2 drift
`predictor_target_to_nn_cos_ratio_at_max_std`	1.51e-4	3.54e-6	2.67e-5	3.39e-6	max std target/NN ratio
Latent noise					
`cka_linear_at_max_std`	0.1986	0.5536	0.3085	0.1814	CKA unperturbed vs noisy
`latent_cost_surface_slope_z`	635.31	1.3886	599.45	0.6208	goal latent cost slope

C Heteroscedastic-loss formulation

The scale-preserving heteroscedastic NLL referenced in Section 6.6 (the "Additional ablation" paragraph) and detailed in Appendix D is:

$$\mathcal{L}_{\text{hetero}} = \frac{1}{2}\,\exp(-s_t)\,\lVert z_{t+1} - \hat{z}_{t+1}\rVert^2 + \frac{1}{2}\,s_t\,, \tag{8}$$

where s_t is the σ-head-predicted log-variance. During training $\exp(-s_t)$ acts as an automatic weight: high-error transitions are down-weighted and low-error transitions are up-weighted. When $s_t \equiv 0$ the loss reduces to plain MSE.

D Heteroscedastic-loss negative result (data)

This appendix records the full data behind the "Additional ablation" paragraph in Section 6.6. The σ-head is trained jointly with the predictor; gradient is detached on the σ path so the mean prediction path is exactly LeWM MSE when σ is constant.

Table 13: Heteroscedastic-loss evaluation.

Task / model	Unperturbed	goal 0.05	pixels 0.05	px+goal 0.05	goal 0.08	px+goal 0.08
TwoRoom LeWM-base	94.00	73.33	72.33	61.33	58.67	50.00
TwoRoom LeWM+noise point-best	98.33	98.00	98.33	98.00	98.67	98.67
TwoRoom hetero	99.67	85.33	96.67	84.67	73.33	55.33
PushT LeWM-base	86.33	38.00	17.00	12.00	11.33	4.67
PushT LeWM+noise point-best	89.33	87.67	88.00	88.33	89.67	87.00
PushT hetero	13.33	7.67	7.67	7.67	9.67	6.00

TwoRoom hetero reaches 99.67% on unperturbed evaluation — consistent with the prior that low-dimensional discrete tasks benefit from stronger nuisance contraction / clustering — but lags noise training on high-noise robustness. **PushT hetero unperturbed success is** 13.33% **— a method-level failure**, not evidence for a useful robustness compromise.

Table 14: Representation diagnostics of the hetero-loss failure.

Metric	TwoRoom base	TwoRoom hetero	PushT base	PushT hetero
`clean_nn_cos_dist_median`	0.0449	0.0281	0.2360	0.1051
`clean_effective_rank`	47.60	33.59	76.42	42.85
`transition_resolution_ratio_`_cos	0.5538	0.3780	0.0868	0.0101
`transition_resolution_ratio_`_l2	0.7216	0.6055	0.3015	0.1023
`id_probe_r2`	0.2889	−0.0573	0.7739	0.2678
`action_mean_pred_shift_norm`	0.5329	0.4482	0.1283	0.0841
`predictor_rollout_T8_l2`	18.62	17.90	18.65	14.01

Hetero-loss compresses the representation in both tasks. For TwoRoom (low-dimensional, discrete, redundant), compression is acceptable. For PushT, the L2 transition-resolution ratio collapses from 0.3015 to 0.1023 and `id_probe_r`^{2} from 0.7739 to 0.2678 — task-relevant state information is erased. The drop in multi-step rollout drift is not good news either: the latent has become more *predictable* without being more *controllable*.

Mechanism. The σ-head correctly learns per-transition difficulty (high σ on contact frames in PushT, on doorway crossings in TwoRoom, etc.), but using σ as an automatic weight to down-weight high-error transitions misclassifies the contact-and-control-critical states of PushT as "unimportant" and erases them. This is a direct demonstration of the broader selective-consistency lesson: in contact-heavy control, **hard $\neq$ unimportant**.

This finding motivates a follow-up direction we are currently exploring: per-token *consistency* (not loss-reweighting) controllers driven by detached difficulty signals, which preserve the mean prediction path's gradient distribution. That work is independent of this paper's diagnostic study and will be reported separately.

E One-off cost-swap sanity check

This appendix records the eval-only cost-swap ablation referenced in Section 5.8.1. It is **not** part of the 36-checkpoint canonical evaluation table and uses a separate unperturbed reference (num_eval = 300), so we report it only as a sanity check rather than a pooled statistic.

Table 15: One-off eval-only cost swap on a TwoRoom checkpoint, std = 0.03 pixels+goal. Reference row reports a separate unperturbed eval of the same checkpoint (num_eval = 300).

Variant	Cost type	Cost space	Success rate
A (default)	cosine	normalized	36.0
B (swap)	mse	raw	42.0
Unperturbed reference (same ckpt)	—	—	69.7

Swapping cost recovers only +6 pt (36 → 42), still far below the unperturbed reference (69.7). The narrow reading is therefore: **a sanity check suggests the cost function alone is unlikely to explain the visual-corruption collapse.**

F Cross-method replication: PLDM noise sweep on four tasks

This appendix records the second method-family on which the noise sweep and the cross-checkpoint correlation analysis replicate. PLDM follows the joint-embedding predictive line from Sobal et al. [29]; the concrete latent-dynamics planning baseline is from Sobal et al. [30]. We use the implementation distributed through the `stable-worldmodel` baseline suite [22]. Training and evaluation follow the LeWM canonical protocol bit-for-bit: per-frame Gaussian pixel noise via `utils.py::AddNormalizedGaussianNoise`, `std_max` $\in \{0.01, \ldots, 0.08\}$, three evaluation seeds (42/43/44), 100 trajectories per seed.

Coverage. The PLDM release is complete for the same grid as the LeWM canonical sweep: 4 tasks × 9 configurations (no-noise baseline plus `std_max` $\in \{0.01, \ldots, 0.08\}$), three evaluation seeds (42/43/44), and 100 trajectories per seed. The PLDM eval, predictor-diagnostic, full-diagnostic, and cross-method-correlation JSON files are listed in the data manifest; we do *not* merge them into the LeWM canonical files used by Tables 1–9.

No-noise-trained PLDM cliff. PLDM reproduces the no-noise-trained visual-corruption cliff on PushT and TwoRoom, while Reacher and Cube show much smaller drops under this PLDM training recipe (Table 16). This supports a method-family reading of the failure mode without claiming every task has the same cliff magnitude.

Table 16: PLDM baseline trained without input-noise augmentation on all four tasks. Success rate mean ± population std, 3 seeds × 100 trajectories.

Task	unperturbed	pixels 0.05	pixels 0.08	px+goal 0.08	unperturbed → pixels 0.08 drop
TwoRoom	96.67 ± 1.70	79.67 ± 4.03	67.00 ± 2.16	51.67 ± 2.05	-29.67
PushT	75.33 ± 3.68	46.00 ± 4.97	18.33 ± 2.05	10.00 ± 2.16	-57.00
Reacher	82.67 ± 1.70	81.33 ± 4.03	80.33 ± 5.73	79.33 ± 0.94	-2.33
Cube	52.67 ± 3.30	50.33 ± 4.50	48.33 ± 4.50	48.33 ± 2.87	-4.33

PLDM full sweep — unperturbed evaluation. Table 17 reports the per-task unperturbed success rate at every trained `std_max` level. Table 18 reports the corresponding pixels 0.08 robustness with an unperturbed goal.

Table 17: PLDM+noise sweep, unperturbed success rate (%; released condition-name: `clean`). † marks the per-task unperturbed point-best.

`std_max`	TwoRoom	PushT	Reacher	Cube
0 (base)	96.67 ± 1.70	75.33 ± 3.68	82.67 ± 1.70	52.67 ± 3.30
0.01	96.33 ± 0.94	72.33 ± 4.19	78.00 ± 0.82	51.33 ± 1.25
0.02	97.33 ± 0.47	71.33 ± 3.86	82.33 ± 2.36	53.33 ± 1.70
0.03	96.67 ± 1.70	$76.67 \pm 7.54^{\dagger}$	83.00 ± 3.74	52.67 ± 3.30
0.04	97.67 ± 0.47	68.67 ± 5.25	$85.00 \pm 2.16^{\dagger}$	54.00 ± 2.94
0.05	97.67 ± 1.25	74.33 ± 3.40	72.00 ± 1.41	54.00 ± 2.16
0.06	98.00 ± 0.82	70.33 ± 6.18	79.00 ± 0.82	$56.00 \pm 4.97^{\dagger}$
0.07	98.00 ± 0.82	66.67 ± 8.38	80.67 ± 2.62	53.67 ± 3.68
0.08	$98.33 \pm 0.94^{\dagger}$	68.00 ± 1.41	80.33 ± 1.25	54.00 ± 1.63

Table 18: PLDM+noise sweep, pixels 0.08 success rate with unperturbed goal (%). ‡ marks the per-task pixels 0.08 point-best.

`std_max`	TwoRoom	PushT	Reacher	Cube
0 (base)	67.00 ± 2.16	18.33 ± 2.05	80.33 ± 5.73	48.33 ± 4.50
0.01	96.00 ± 1.63	23.33 ± 2.49	77.67 ± 3.40	51.33 ± 2.36
0.02	95.67 ± 0.94	47.00 ± 0.82	80.67 ± 5.25	51.00 ± 4.32
0.03	96.33 ± 1.25	$73.00 \pm 7.48^{\ddagger}$	$81.67 \pm 4.71^{\ddagger}$	54.33 ± 2.62
0.04	97.33 ± 0.94	66.67 ± 5.73	80.33 ± 2.87	$54.67 \pm 2.36^{\ddagger}$
0.05	97.67 ± 1.89	71.67 ± 5.79	62.67 ± 4.19	54.00 ± 3.74
0.06	$98.00 \pm 0.00^{\ddagger}$	69.33 ± 4.78	75.00 ± 0.82	54.33 ± 0.94
0.07	98.00 ± 0.82	64.00 ± 7.79	78.33 ± 2.05	53.00 ± 3.74
0.08	97.67 ± 1.25	68.00 ± 5.66	79.33 ± 1.25	53.67 ± 2.05

Reading. Four quantitative observations are useful for interpreting PLDM as a second method family:

1. **TwoRoom (visually redundant) benefits from heavy noise on both methods.** PLDM pixels 0.08 success rises from 67.00% at the no-noise baseline to 98.00% at `std_max` $= 0.06$, then remains on a high-noise plateau. This mirrors the LeWM reading that TwoRoom tolerates strong same-state nuisance contraction.
2. **PushT (contact-heavy) exhibits the corruption cliff and large noise-training recovery.** PLDM without noise augmentation drops by 57.00 pt under pixels 0.08; training with `std_max` $=$ 0.03 recovers pixels 0.08 success to 73.00% (+54.67 pt over the no-noise baseline). The unperturbed and pixels 0.08 point-bests are co-located at `std_max` $= 0.03$ on PLDM.
3. **Reacher is already robust under no-noise-trained PLDM.** The no-noise PLDM drop is only 2.33 pt, and the observation-noise point-best sits at `std_max` $= 0.03$ (81.67%). We therefore treat Reacher as a low-cliff external baseline rather than evidence for a universal noise-training gain.
4. **Cube (structured 3D manipulation) remains weakly noise-responsive.** PLDM Cube unperturbed success spans 51.33–56.00% and pixels 0.08 spans 48.33–54.67% across the full grid. This is the same weak noise responsiveness seen on LeWM.

Method-specific details. PLDM's PushT unperturbed point-best (76.67% at `std_max` $= 0.03$) is lower than LeWM's (89.67% at `std_max` $= 0.03$), so the external baseline is not a performance leaderboard. It is a robustness check: the cliff-and-recovery shape is reproduced, while the exact unperturbed/robust optimum location is method-dependent. In particular, the LeWM PushT optimum dissociation should be read as a LeWM-family signature, not a cross-method law.

Cross-method partial correlations. We repeat the fragility ratio partial-correlation analysis for PLDM on all four tasks. The headline PushT null replicates in the limited sense that we do not see stable evidence for a non-zero residual corruption-gap association: PLDM has unconditional $\rho(\text{metric}, \text{corruption drop}) = -0.75$ but partial $\rho(\text{metric}, \text{corruption drop}) \mid \texttt{std\_max} = -0.05$ (95% bootstrap CI $[-0.92, +0.61]$), and the joint LeWM+PLDM PushT partial after conditioning on `std_max` and method is $+0.22$ (95% CI $[-0.59, +0.61]$). The intervals are wide and include zero. Other tasks show task-specific residuals, so we use the full table as a scope boundary rather than as a universal diagnostic claim.

Table 19: PLDM fragility ratio partial correlations, $n = 9$ per task. The first three partial columns condition on `std_max` within PLDM; the final column uses the joint LeWM+PLDM sample ($n = 18$) and conditions on both `std_max` and a method dummy. Numbers come from the released PLDM correlation JSON.

Task	PLDM n	unperturbed partial	pixels 0.08 partial	corruption-drop partial	joint corruption-drop partial
TwoRoom	9	−0.26	−0.43	+0.35	+0.12
PushT	9	−0.22	−0.13	−0.05	+0.22
Reacher	9	−0.68	−0.49	+0.17	+0.43
Cube	9	−0.36	+0.19	−0.53	−0.13

PLDM full-diagnostic mechanism check. We also run the same five diagnostic layers on the PLDM sweep and release the resulting per-checkpoint summary in `canonical_full_diagnostics_pldm_20260523.json`. The compact base-vs-representative view in Table 20 shows a different mechanism profile from the LeWM compression chain in Table 6: PLDM's representative recovery checkpoints do *not* exhibit a broad collapse in rank, transition resolution, or inverse-dynamics probe. The most consistent movement is instead a reduction in multi-step predictor drift. This is useful precisely because it separates two claims: the task-level fragility/recovery signature replicates across method families, while the exact diagnostic mechanism is architecture-dependent.

Table 20: PLDM full-diagnostic check, no-noise baseline → representative noise-trained checkpoint. Representatives are each task's PLDM pixels 0.08 point-best ckpt (TwoRoom `std_max` = 0.06, PushT `std_max` = 0.03, Reacher `std_max` = 0.03, Cube `std_max` = 0.04). Values come from the released full-diagnostics JSON.

Task	representative `std_max`	effective rank	transition ratio L2	inverse-dynamics probe R^2	rollout T8 L2
TwoRoom	0.06	74.78 → 72.70	0.8188 → 0.8254	0.3233 → 0.3201	20.36 → 1.02
PushT	0.03	130.13 → 131.90	0.4710 → 0.4778	0.7570 → 0.7479	20.25 → 2.82
Reacher	0.03	117.52 → 108.14	0.5053 → 0.4824	0.2150 → 0.1844	1.45 → 0.94
Cube	0.04	134.80 → 124.49	0.6395 → 0.6313	0.5428 → 0.5576	18.98 → 4.02

What this strengthens, and what it does not. PLDM strengthens C2 by showing that the four task signatures are not tied to one LeWM checkpoint family. It strengthens C4 only in the precise PushT sense stated in the main text: after removing the sweep-level `std_max` effect and the method offset, the local predictor-sensitivity diagnostic does not isolate corruption-specific robustness. The TwoRoom and Cube PLDM residuals are deliberately left visible in Table 19. They do not contradict the headline null: the intervals are wide, the noisy-eval ranges are small on the saturated tasks, and the residuals should not be converted into a robustness oracle.

Mechanism boundary. Table 20 is the reason we do not promote the LeWM compression chain to a universal theorem. On LeWM, noise-trained representatives can compress effective rank and reduce transition-key resolution; on PLDM, the same task-level robustness pattern is accompanied mainly by reduced rollout drift with largely preserved rank/resolution probes. The shared conclusion is therefore about *control-time visual fragility and noise-training recovery*; the internal diagnostic route is method-specific.

G Cross-corruption sanity check: no-noise-trained blur evaluation

This appendix addresses the narrow concern that the observed visual fragility is an artefact of Gaussian noise only. We evaluate the LeWM and PLDM baselines trained without input-noise augmentation under Gaussian blur with kernel sizes $3, 7, 11, 15$, applied to pixels, goal images, or both. This is an **eval-only** stress test: no blur-trained checkpoint is included, and the blur data are not mixed into the Gaussian-noise sweep. Source: `canonical_blur_baselines_20260523.json`.

Table 21: Pixels+goal blur stress test on baselines trained without input-noise augmentation. Success rate mean $\pm$ population std over 3 evaluation seeds $\times$ 100 trajectories. Kernel sizes 11 and 15 are severe low-pass endpoints on 224×224 inputs, so they should not be read as mild corruptions.

Method	Task	unperturbed	$k = 3$	$k = 7$	$k = 11$	$k = 15$	worst drop
LeWM	TwoRoom	94.00 ± 3.56	39.33 ± 2.05	32.67 ± 3.09	30.33 ± 2.87	30.67 ± 0.47	-63.67
LeWM	PushT	86.33 ± 2.36	81.67 ± 4.50	74.00 ± 1.41	65.33 ± 2.49	58.67 ± 6.65	-27.67
LeWM	Reacher	58.67 ± 1.25	57.00 ± 6.38	45.67 ± 2.05	36.00 ± 4.90	20.33 ± 2.49	-38.33
LeWM	Cube	66.67 ± 2.62	65.00 ± 1.63	62.33 ± 2.62	57.33 ± 2.87	54.00 ± 2.16	-12.67
PLDM	TwoRoom	96.67 ± 1.70	38.33 ± 0.47	30.33 ± 2.87	28.33 ± 1.25	28.33 ± 1.70	-68.33
PLDM	PushT	75.33 ± 3.68	74.00 ± 2.94	72.00 ± 3.56	67.67 ± 4.03	62.33 ± 2.05	-13.00
PLDM	Reacher	82.67 ± 1.70	86.00 ± 2.16	80.00 ± 3.56	72.00 ± 2.16	68.00 ± 4.08	-14.67
PLDM	Cube	52.67 ± 3.30	53.00 ± 5.35	53.00 ± 2.83	50.67 ± 3.68	52.33 ± 5.44	-2.00

Reading. Blur is a different failure mode from Gaussian pixel noise. The clear collapse is TwoRoom: it falls below 40% already at $k = 3$ on both LeWM and PLDM and then saturates around 28–33% for $k = 7$–15. This reverses the Gaussian-noise ordering, where PushT is the most fragile task. The other tasks are more stable under blur: PLDM PushT, PLDM Reacher, and both Cube baselines lose at most about 15 pt at the strongest endpoint, while LeWM PushT degrades gradually and LeWM Reacher drops sharply only at $k = 15$. We therefore use blur only as an eval-only cross-corruption sanity check: visual fragility is not unique to Gaussian noise, but the task ordering and recovery profile are corruption-specific.

H Phase-0 paired ACPC diagnostics

This appendix records the first full-coverage run of the paired action-conditioned diagnostics defined in Section 3.2. The artifact `assets/paper1_data/acpc_phase0_diagnostics.json` covers 72 checkpoint rows: LeWM and PLDM, four tasks, and nine training-noise levels per task. All rows completed successfully. Each row uses 100 sampled sequences, Gaussian pixels+goal corruption at std 0.08, rollout horizon $H = 8$, and a shared candidate-action set of one dataset future plus 64 random futures. Because this uses the auxiliary pixels+goal stress endpoint rather than the paper's primary unperturbed-goal endpoint, it is reported as a broader exploratory sanity check rather than as the main ACPC evidence. PCC, CRA, and MAF are computed from the same original/noisy candidate-cost vectors. ADM and SPRR use an action-distance latent proxy, not an oracle state/keypoint margin.

Table 22: Phase-0 paired ACPC diagnostics, no-noise baseline → px+goal 0.08 point-best checkpoint. Lower is better for ACPC-H/transition, PCC, and MAF; higher is better for CRA and top-8 overlap. Values are exploratory diagnostics, not predictor-selection rules.

Task	Method	robust `std_max`	px+goal 0.08	ACPC-H/transition	PCC median	CRA mean	top-8 overlap	MAF flip
TwoRoom	LeWM	0.08	50.00 → 98.67	1.254 → 0.042	176.6 → 7.0	0.157 → 0.990	0.201 → 0.952	0.92 → 0.06
TwoRoom	PLDM	0.05	51.67 → 98.33	1.062 → 0.063	164.3 → 7.1	0.367 → 0.982	0.287 → 0.911	0.85 → 0.05
PushT	LeWM	0.06	4.67 → 87.00	2.740 → 0.149	67.3 → 4.6	0.323 → 0.986	0.213 → 0.914	0.96 → 0.08
PushT	PLDM	0.03	10.00 → 72.00	1.682 → 0.162	52.0 → 8.1	0.300 → 0.967	0.286 → 0.869	0.90 → 0.08
Reacher	LeWM	0.02	15.00 → 85.67	1.922 → 0.023	528.0 → 1.2	0.242 → 0.999	0.219 → 0.990	0.97 → 0.01
Reacher	PLDM	0.04	79.33 → 81.33	0.104 → 0.063	11.7 → 6.3	0.965 → 0.987	0.930 → 0.949	0.13 → 0.10
Cube	LeWM	0.07	46.33 → 68.00	1.944 → 0.055	88.8 → 4.1	0.337 → 0.996	0.266 → 0.951	0.88 → 0.00
Cube	PLDM	0.08	48.33 → 56.33	1.136 → 0.034	113.6 → 3.0	0.663 → 0.997	0.512 → 0.967	0.74 → 0.01

Reading. The Phase-0 pass is useful because it converts the ACPC definitions from a purely prospective protocol into a release artifact that behaves sensibly on the existing sweep. The largest Gaussian-noise cliffs (TwoRoom and PushT on both methods, Reacher/Cube on LeWM) coincide with large baseline ACPC-H, PCC, ranking disruption, and action flips, and the px+goal point-best checkpoints reduce these quantities sharply. PLDM Reacher is the important scope boundary: it already has high px+goal success at the no-noise baseline, and the paired diagnostics are correspondingly already close to the recovered checkpoint. Across the joint LeWM+PLDM sample within each task, partial Spearman correlations after conditioning on `std_max` and method remain task-specific: ACPC-H/transition vs. corruption drop is strong on PushT (+0.67) and Reacher (+0.71), but only modest on TwoRoom (+0.21) and Cube (+0.33). We therefore treat Table 22 as a face-validity and mechanism-localisation check, not as evidence that ACPC-H, PCC, CRA, MAF, ADM, or SPRR can by themselves predict robustness.